Module 3: Binomial and Poisson Models — Practice

Lesson 4 | Calculate, justify, and interpret

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Preparation

Read Lesson 4. Use the Martin Clothing Store and arrivals inputs already introduced in English and Arabic. These are model calculations rather than an analysis of newly supplied observations.

1. Select the model

Explain why the store example can be binomial with $n = 3$ and $p = 0.30$, and why the arrival count can be Poisson with mean 10 per 15 minutes. Give one plausible way each model's assumptions could fail.

2. Translate binomial events

Calculate the probability of exactly two, at most two, at least two, and more than two purchases. Then find the expected count, variance, and SD. Explain the difference between `dbinom()` and `pbinom()` and the boundary needed for "at least two."

3. Change the exposure

For the same stated arrival rate, find:

1. The mean count in three minutes.
2. The probability of no arrivals in three minutes.
3. The probability of at least two arrivals in three minutes.
4. The probability of more than ten arrivals in fifteen minutes.

Explain why the model mean must be tied to the interval and why the count has no fixed upper bound in the Poisson model.

4. Simulate and interpret

Use a fixed seed and simulate 10,000 three-customer binomial counts. Estimate the probability of at least two purchases and compare it with the exact value. Explain why a different seed may change the estimate and why this comparison does not establish model fit for real customers.

5. Sampling without replacement

Explain why taking three records from a finite list without replacement need not give independent trials with the same success probability. Do not apply a binomial model merely because there are three selections and two possible labels.

Submit a method-and-event table with results and a short interpretation. Return to the lesson.